

Phase 5 – Project Development Phase

PROJECT: EduGenie – Google Gemini Powered Learning Assistant

OBJECTIVE

To develop the EduGenie backend, AI modules, frontend, and live integration.

BACKEND DEVELOPMENT

FastAPI is used to create RESTful endpoints for each feature.

IMPLEMENTED MODULES

1. Explanation Module

Uses LaMini-Flan-T5-783M to provide concise and easy-to-understand explanations.

2. QnA Module

Uses Gemini to answer academic and general knowledge questions.

3. Quiz Module

Creates structured quiz questions with four options and a correct answer.

4. Summary Module

Uses Gemini to convert long paragraphs into concise, easy-to-understand summaries.

5. Learning Path Module

Uses Gemini to generate beginner-to-advanced learning paths with useful learning resources.

FRONTEND DEVELOPMENT

- Created index.html.
- Added task selection for Explain, QnA, Quiz, Summary, and Recommend Path.
- Added text input area.
- Added submit button.
- Added responsive CSS styling.
- Added result display section.

LIVE INTEGRATION

The frontend sends a POST request to the FastAPI backend, and the generated result is displayed below the input area.

PROCEDURE

- 1 Set up the Python environment.
- 2 Install required packages.
- 3 Configure the Gemini API.
- 4 Develop the AI modules.
- 5 Create FastAPI endpoints.
- 6 Develop the HTML/CSS frontend.
- 7 Connect frontend forms to backend APIs.
- 8 Run the application locally and verify the integration.